

Hydroxy compounds

A Level Chemistry

Alcohols

An **alcohol** 醇 has the -OH (hydroxyl) functional group.

Making alcohols

Method	Reagents and conditions
addition of steam to an alkene	$\text{H}_2\text{O(g)}$, H_3PO_4 catalyst (electrophilic addition 亲电加成)
an alkene 烯烃 with cold dilute KMnO_4	gives a diol 二醇 (two -OH groups)
substitution of a halogenoalkane 卤代烷	NaOH(aq) , heat
reduction 还原 of an aldehyde 醛 or ketone 酮	NaBH_4 or LiAlH_4
reduction of a carboxylic acid 羧酸	LiAlH_4
hydrolysis 水解 of an ester 酯	dilute acid or alkali, heat

Reactions of alcohols

- **combustion**: alcohols burn in oxygen to give carbon dioxide and water.
- **substitution** to a halogenoalkane, for example with HX , PCl_5 , PCl_3 and heat, or SOCl_2 .
- **with sodium**: alcohols react with sodium metal to give hydrogen and a sodium alkoxide —like water, but more slowly.
- **oxidation** 氧化 with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ (or KMnO_4). The product depends on the class of alcohol (see below).
- **dehydration** 脱水 to an alkene, using a hot Al_2O_3 catalyst or concentrated acid.
- **ester formation**: an alcohol reacts with a carboxylic acid (with concentrated H_2SO_4 catalyst) to make an ester.

Three classes and how oxidation tells them apart

An alcohol is **primary** 伯, **secondary** 仲 or **tertiary** 叔, depending on how many carbons are joined to the carbon holding the -OH . Some molecules have more than one -OH group.

Class	Oxidation product
primary	aldehyde (by distillation 蒸馏), then carboxylic acid (by reflux 回流)
secondary	a ketone
tertiary	not oxidised

In a quick test, acidified $\text{K}_2\text{Cr}_2\text{O}_7$ turns from **orange to green** with a primary or secondary alcohol, but stays orange with a tertiary alcohol.

- **distillation** removes the aldehyde as it forms, before it can be oxidised further.
- **reflux** keeps boiling the mixture and returning the vapour, so the alcohol is fully oxidised to the carboxylic acid.

The iodoform test

If you warm an alcohol that contains the $\text{CH}_3\text{CH(OH)-}$ group with alkaline aqueous iodine, you get a pale yellow precipitate of **tri-iodomethane** 三碘甲烷 (CHI_3) and the ion RCO_2^- . This is a useful test for that group.

Acidity of alcohols

The -OH group makes alcohols very weakly acidic: they can lose the H^+ to form an RO^- ion. But their **acidity** 酸性 is **lower** than that of water. This is because the alkyl group pushes electron density onto the oxygen, which makes the RO^- ion less stable, so the alcohol holds onto its H^+ more tightly.